

Clinical Study Summary Report (CSSR)

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1. Administrative & Study Identification

Full Study Title: An Observational Multi-Country Post-Authorisation Safety Study to Evaluate the Risk of Serious Adverse Cardiovascular Events in Adolescent and Adult Patients With Severe Asthma Taking Tezepelumab

Short Title/Acronym: Tezspire Cardiac Events PASS

Protocol Number: D5180R00024

NCT Number: NCT06951867

Sponsor Name: AstraZeneca

Company: AstraZeneca

Investigational Product: Tezepelumab (TEZSPIRE®)

Phase of Development: Not Applicable

Trial Status: RECRUITING

Medical Monitor: AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the risk of serious adverse cardiovascular events in adolescent and adult patients with severe asthma taking tezepelumab compared to a population receiving standard of care (SOC) treatment for severe asthma.

Secondary Objectives: To compare long-term cardiovascular safety data across multiple real-world data sources to inform the European Union risk management plan.

2.2 Background & Rationale Results from DESTINATION, a long-term extension, phase 3 trial, showed a numeric imbalance in the occurrence of cardiac disorder system organ class serious adverse events (SAEs) in participants receiving tezepelumab compared to placebo. The observed imbalance resulted in the inclusion of serious cardiovascular events as an important potential risk to be addressed in the European Union (EU) risk management plan (RMP). AstraZeneca proposed to evaluate the risk of serious cardiovascular events with long-term tezepelumab treatment to meet regulatory requirements under a category 3 Post-Authorisation Safety Study (PASS).

3. Study Design & Methodology

3.1 Design Framework Study Type: Observational (non-interventional, longitudinal, population-based cohort study)

Control Type: Active Comparator (Unexposed patients treated with non-biologic high-intensity SOC)

Blinding: Open-label (Secondary data analysis)

Randomization: N/A

3.2 Planned Enrollment & Duration Targeted Enrollment: 95,574 patients

Number of Sites: Multiple data sources across Denmark, France, Germany, and the United States of America (USA)

Estimated Study Start: Q1 2022 - Q3 2023 (corresponds to market launch dates)

Estimated Primary Completion: 28-Feb-2029

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Patients with a diagnosis of asthma receiving tezepelumab or high-intensity SOC treatment for severe asthma at any point during the study inclusion period.

4.2 Exclusion Criteria 1. < 12 months of data availability prior to index date.

2. Age < 12 years at index date.

3. History of congenital heart disease or heart transplant.

4. Exposure to non-tezepelumab biologics on index date or within the 5-half-life clearance period of the biologic (for comparative analysis).

5. Presence of non-fatal myocardial infarction or stroke and the specific outcome of interest in the 180 days prior to index date.

6. Matching criteria (including type and duration of SOC exposure in the 12 months before index date and propensity score [PS] matching) will be applied to ensure exposed and unexposed patients' comparability.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Routine clinical practice administration.

Route of Administration: Subcutaneous (Tezepelumab) or per SOC guidelines.

Duration of Treatment: Up to ~5-year study period in each country.

5.2 Concomitant & Prohibited Therapy Permitted: Standard of Care (SOC) for severe asthma.

Prohibited: Non-tezepelumab biologics on the index date or within the 5-half-life clearance period are excluded for comparative cohorts.

6. Study Timeline & Visit Schedule

(Note: As an observational registry/database study, there are no physical patient visits. The timeline aligns with secondary data extraction).

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -365 to 0 Retrospective data query for baseline characteristics and history Baseline Day 0 (Index) Trigger exposure to Tezepelumab or high-intensity SOC
Interim	Annual	Periodic Safety Update Reports (PSURs)	
End of Study	28 February 2029	Final outcome measurement and extraction of cardiovascular events	

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Composite outcome MACE, consisting of non-fatal myocardial infarction, non-fatal stroke, or cardiovascular death.

Measurement Method: Aggregated data extraction from electronic health records (EHR) and claims databases.

7.2 Secondary & Exploratory Endpoints * Composite of four serious adverse cardiovascular events * General cardiac disorder system organ class SAEs * Comparative cardiovascular safety between Tezepelumab and high-intensity non-biologic SOC

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Conducted via retrospective/prospective monitoring of electronic health records and international databases.

Serious Adverse Events (SAEs): Safety signals reported specifically for Major Adverse Cardiovascular Events (MACE).

Data Monitoring Committee (DMC): Pharmacovigilance Risk Assessment Committee (PRAC) monitoring per EMA guidelines.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Exposed cohort (Tezepelumab) and unexposed cohort treated with high-intensity SOC.

Sample Size Justification: Target \$N = 95,574\$ to provide sufficient statistical power for identifying rare cardiovascular SAE signals in real-world data.

Handling of Missing Data: Standard epidemiological imputation techniques for missing claims/EHR parameters where methodologically appropriate.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Pharmacoepidemiology Practices (GPP) and PASS regulatory mandates.

Monitoring Plan: Routine data quality, integrity validation, and signal detection across Denmark, France, Germany, and the USA.

Audit Trail: Robust database logging for tracking baseline inclusion criteria and exposure triggers.

11. Study Identifiers & Governance

Primary ID: D5180R00024

Registry Number: NCT06951867

Sponsor/Lead Organization: AstraZeneca

Study Title: Tezspire Cardiac Events PASS

Official Regulatory Title: An Observational Multi-Country Post-Authorisation Safety Study to Evaluate the Risk of Serious Adverse Cardiovascular Events in Adolescent and Adult Patients With Severe Asthma Taking Tezepelumab

12. Clinical Framework

Disease Areas: Cardiovascular Events, MACE, Severe Asthma

Diagnosis Threshold: Patients ≥ 12 years of age diagnosed with severe asthma receiving targeted treatment

Medical Methodology: Biologic therapy vs. High-intensity standard of care

Optimization Target: Asthma control without elevating risk of serious cardiovascular events

13. Study Procedures & Methodology

Management Pathway: High-intensity standard of care or biologic add-on treatment

Education Protocol (HCP): N/A (Secondary data observational study)

Education Protocol (Patient): N/A (Secondary data observational study)

Quality Audit Mechanism: Data source validation mapping trigger exposures and tracking 12-month retrospective history

14. Observation & Follow-Up Schedule

Total Duration: Up to 5 years (anticipated last date of study period on 28 February 2029)

Onsite Visit Cadence: N/A (Secondary data)

Remote Monitoring Frequency: Continuous pharmacovigilance

tracking

Checklist Review Interval: PSUR safety report intervals (Annual/
Bi-annual)

15. Sign-off & Approval

Role	Name	Signature	Date		----	----	----	----
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					